Chart-Based Risk Engine

Semi-automated execution for SENSEX, NIFTY 50 and BANK NIFTY options on FYERS

Design, feasibility answers, sizing maths, order flow, safety rules and the build plan for the kunal-fyers-terminal project

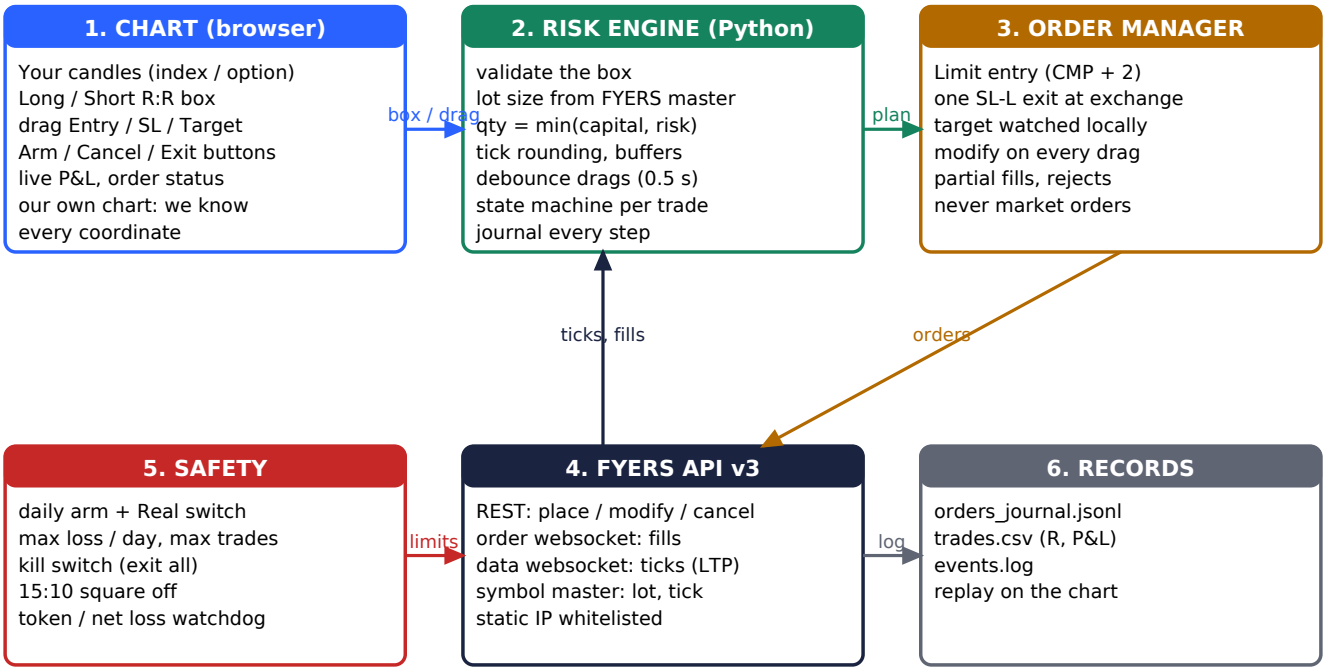
Version 1 - 30 Oct 2026 - for Kunal (owner)

Short answer: YES, it is workable - with one important change. Charts inside FYERS Web / TradingView do not let any program read your hand-drawn Long/Short box or notice when you drag it. So the box must be drawn on our own chart inside kunal-fyers-terminal (the same TradingView-style chart you already use there). On our chart we know every coordinate, so drawing, dragging and order-modify work reliably. Everything else in your specification (dual-constraint sizing, limit + buffer orders, SL-L, dynamic modify) is fully possible with the FYERS API v3.

Where to build it: your suggestion is right - the existing kunal-fyers-terminal repository, on a new branch feature/chart-risk-engine. No new repository is needed: the login, FYERS client, contract list (lot sizes), limit-order executor, daily arm switch, chart and tests are already there and would otherwise be copied.

1. What you asked for, in one picture

You keep doing the analysis. The system only does the arithmetic and the order handling, exactly and every time. It never predicts direction.



How one trade flows:

- You draw a Long box (buy CALL) or a Short box (buy PUT - bearish = buy put, never sell options) on the option chart or the index chart. The box has three handles: Entry, Stop, Target.
- A small ticket appears next to the box: lots, quantity, money used, rupee risk, rupee reward, R:R, and which limit (capital or risk) decided the size. Nothing is sent yet.
- You press Arm. The engine places the BUY LIMIT (entry + Rs 2) when the price is at the entry.
- Once filled, ONE protective SELL SL-L order sits at the exchange (trigger = stop, limit = stop - Rs 2).
- You drag the Stop handle: after you let go, the SL-L order is modified at FYERS (trigger and limit). You drag the Target handle: the engine's target moves (and, if you choose the 'target order at exchange' mode, the order is modified).
- Target reached: the SAME exit order is changed into a SELL LIMIT at target - Rs 2. So there is never a second sell order that could fill twice.

2. Answers to your feasibility questions

2.1 Can FYERS Web / TradingView charts or Pine Script read the Long/Short box?

Way	Can it read the box and its drags?	Verdict
Pine Script (TradingView or FYERS Smart Script)	No. Pine has no access to your manual drawings at all. It only sees prices and its own inputs.	Not possible
Approach A - Chrome extension reading TradingView's page	Only by reaching into TradingView's private code inside the page. Drawings are painted on a canvas; there is no public way to read them. It breaks whenever TradingView or FYERS updates the website (often weekly), and it may break their terms of use.	Fragile - not for real money

Way	Can it read the box and its drags?	Verdict
Approach B - Pine input.price drag points + alert webhooks	You can drag interactive input lines, BUT an alert keeps the values it had when it was created. After a drag you must delete and re-create the alert by hand. Also webhooks need a TradingView paid plan, a public web address for our program, and add seconds of delay. FYERS Smart Script cannot send webhooks to our program.	Workable only for 'set once' levels - not for live dragging
Approach C - separate desktop overlay window	It cannot see the browser chart either; you would type or click levels in a second window that is not lined up with the candles.	Works, but clumsy
Approach D (recommended) - draw the box on OUR chart in kunal-fyers-terminal	Yes, completely. Our chart (TradingView Lightweight Charts) is our own code, so every click, drag and release is an event we receive with the exact price.	Reliable - recommended

You can still look at FYERS Web / TradingView for analysis on a second screen. The trade itself is drawn and managed on the terminal chart, which shows the same FYERS candles (index or option) with your KTMI, VWAP and BBP panes.

2.2 Brackets / OCO: does FYERS do it for us?

FYERS feature	What it does	Use it?
Order types 1 Limit, 4 Stop-Limit (SL-L)	Exactly what your spec needs (entry limit, SL-L exit)	YES - the core
Modify order (price, trigger, qty, type)	Change a pending order without cancelling it	YES - every drag
Order websocket (fills / rejects live)	Tells us at once when an order fills, part-fills or is rejected	YES - drives the state machine
Take-profit / stop-loss offsets on place_order (API v3.2)	FYERS attaches TP/SL legs to an entry. It is not clear yet whether these legs exit with LIMIT or MARKET orders, and whether they work for INTRADAY options.	Test with 1 lot first; use only if the exits are limit orders
GTT / OCO orders	Two-leg trigger orders, documented for CNC / MARGIN / MTF products	Not for intraday option buying
Market orders	-	NEVER (your rule, test-guarded in the code)

Our bracket = 'one exit order' synthetic OCO. Placing BOTH a target sell and a stop sell for the full quantity is dangerous: if the price touches the target and then falls, both could fill and you would be SHORT an option (unlimited risk, margin call) - and FYERS may reject the second sell for margin. So only ONE sell order ever exists: the SL-L protection at the exchange (it protects you even if your PC or internet dies). The engine watches the target with live ticks and, when it is reached, modifies that same order into a limit sell at the target. One order, one fill, no double exit.

3. Position sizing: the two limits

Every trade is sized twice and the smaller quantity wins, always in whole lots.

Inputs

Input	Meaning	Example
Total capital	FYERS Available Balance (real) or your Paper capital	Rs 50,000
Capital allocation %	Most of the capital one trade may use	50 % = Rs 25,000
Max risk per trade	Most rupees one trade may lose	Rs 5,000
Entry, Stop, Target	From the box handles (option premium)	100 / 75 / 150
Lot size, tick size	From the FYERS contract list every morning (not typed)	SENSEX 20, tick 0.05
Buffer	Limit buffer for entry and stop (setting)	Rs 2

Formulas

```
Capital-capped: lots_capital = floor( capital x allocation% / (entry_limit x lot) )
Risk-capped: lots_risk = floor( max_risk / (risk_per_unit x lot) )
Final: lots = min(lots_capital, lots_risk); quantity = lots x lot. Zero lots = no trade, with
the reason on the ticket.
```

Your example, done two ways

	Your spec (plain)	Strict (recommended)
Entry price used	100	102 (the limit we send: worst fill)
Stop price used	75	73 (SL-L limit: worst normal fill)
Risk per unit	25	29
Capital check	25,000 / (100 x 20) = 12.5 -> 12 lots (240)	25,000 / (102 x 20) = 12.25 -> 12 lots
Risk check	5,000 / (25 x 20) = 10 -> 10 lots (200)	5,000 / (29 x 20) = 8.6 -> 8 lots (160)
Final	10 lots = 200 qty, worst real loss about Rs 5,800 + charges	8 lots = 160 qty, worst normal loss Rs 4,640 + charges (under Rs 5,000)

Why strict: with buffers, the entry can fill up to Rs 2 higher and the stop can fill up to Rs 2 lower, so 'plain' sizing can quietly lose more than your Rs 5,000 cap. Strict sizing also adds the estimated charges (brokerage, STT, exchange, GST, stamp - about Rs 60-120 a trade) to the risk. You choose: a 'Strict sizing' switch, ON by default.

Extra checks before any order

- Long box: stop below entry and target above entry; Short box (buy PUT) the same on the PUT premium (drawn on the index chart it is mirrored - see section 6).
- Quantity above the exchange freeze limit for one order: split into several orders (FYERS slice orders) - the limits are a setting, checked against the exchange circular.
- Stop too close (less than 3 ticks) or R:R below a minimum you set: warning, Arm stays disabled.
- Daily loss limit reached, max trades per day reached, or after 15:00: Arm disabled.
- Prices rounded to the 0.05 tick: buy limits round UP, sell limits round DOWN (already in the code).

4. Orders: exactly what is sent to FYERS

Step	FYERS order	Example (entry 100, stop 80, target 150, buffer 2)
Entry	type 1 LIMIT, side BUY, productType INTRADAY, validity DAY	limitPrice 102.00. Not filled in 10 s: re-priced once to LTP + 2, at most Rs 1 above the first limit (your existing rule). Still not filled: cancelled, ticket says why.
Protection	type 4 SL-L, side SELL, qty = FILLED qty	stopPrice 80.00, limitPrice 78.00 (placed the moment the entry fills; part fill -> SL for the part, raised as more fills)
Stop dragged to 85	modify_order(id, stopPrice, limitPrice)	stopPrice 85.00, limitPrice 83.00
Target reached (LTP >= 150)	modify the SAME order: type 1 LIMIT SELL	limitPrice = min(150, LTP) - 2 = 148.00 -> fills at the best bid (often 150+)
Target dragged to 160	engine target moves (no order yet)	the ticket shows the new R:R
You press Exit now	modify the SAME order: LIMIT SELL at LTP - 2	never market
15:10	same as Exit now, for every open trade	your daily rule

Optional 'target order at the exchange' mode

If you prefer to SEE the target order in FYERS: the engine keeps a LIMIT SELL at the target and NO stop order at the exchange; the stop is then watched by the engine (ticks) and executed as a limit sell at stop - 2. Faster target fills, but if the PC / internet dies there is no stop at the exchange. Default stays 'stop at exchange' (safer).

Why SL-L can still fail - and what the engine does

In a very fast fall the price can jump from 80 to 74 in one tick. The SL-L triggers at 80 but its limit 78 is above the market, so it does not fill. Your rule forbids market orders, so the engine 'chases': every 2 seconds it lowers the limit to LTP - 2, up to a maximum number of steps, with a red alert and a sound. This is the honest risk of limit-only exits; the backtest already models it.

5. Dragging the box: how a modify is sent

- While you drag, only the drawing and the ticket move (no orders). The order is modified when you RELEASE the handle, and only if the price really changed by at least one tick.
- Debounce 0.5 s and at most a few modifies per second (FYERS allows about 10 requests a second); a drag storm is merged into one modify.
- Rules on release: a stop dragged above the current price (long) would trigger at once -> the engine asks 'Exit now?' instead of sending an instantly-triggered order. Stop dragged further away -> the risk grows: if the new rupee risk is above Max risk, the drag is refused (or quantity reduction is offered, your choice). Quantity is never increased by a drag.
- Every modify is confirmed by the order websocket; the handle turns grey while pending and back to colour when FYERS confirms. If FYERS rejects, the handle jumps back to the price FYERS actually holds.
- The line on the chart always shows the price FYERS holds - never a hoped-for price.

6. Box on the option chart or on the index chart

	Box on the OPTION chart (recommended first)	Box on the INDEX chart
Levels mean	option premium (exact)	index points (your S/R zones)

	Box on the OPTION chart (recommended first)	Box on the INDEX chart
Stop at exchange	SL-L on the option at your stop	engine watches the INDEX; when the index crosses your stop it sells the option with a limit. Plus a wide 'disaster' SL-L on the option at the exchange in case the PC dies
Risk in rupees	exact: (entry - stop) x qty	estimated with the option delta (changes during the trade) - sized with a safety margin
Which strike	the one on the chart	rule you choose: ATM / ATM+1 / premium range / per lot amount (already built)
Short box	draw it on the PUT chart as a Long box	Short box on the index = buy the PUT

Plan: phase 1 = option chart only (exact risk). Phase 2 = index chart with the mapping above.

7. The trade life cycle (state machine)

Every trade is always in exactly one state, saved to disk after every change, so a restart picks up exactly where it was (and re-reads the real orders from FYERS before doing anything).

State	What is at the exchange	Moves to	Trigger
DRAFT	nothing (box drawn, not armed)	ARMED	you press Arm (or auto-arm setting)
ARMED	nothing; engine watches entry condition	ENTRY_WORKING	price reaches entry (or at once if 'enter now')
ENTRY_WORKING	BUY LIMIT at entry + 2	IN_TRADE / CANCELLED	filled (full or part) / not filled in N s
IN_TRADE	ONE SELL SL-L: trigger = SL, limit = SL - 2, qty = filled qty	TARGET_EXITING / STOPPED / EXITING	LTP >= target / SL order filled / opposite action
TARGET_EXITING	the same order modified to SELL LIMIT at target - 2 (or LTP - 2)	CLOSED	filled; if not filled in N s: re-price down, never market
STOP_CHASING	SL triggered but price gapped below the limit	CLOSED	engine lowers the limit step by step (alert on screen)
CLOSED	nothing	-	trade written to trades.csv

8. Safety rules and edge cases

Situation	What the system does
PC, browser or internet dies	The SL-L protection is AT THE EXCHANGE, so the stop still works. On restart the engine reads the orderbook + positions from FYERS and rebuilds each trade.
FYERS token expires / logged out	No new entries; open trades keep their exchange SL; red banner 'log in again'.
Partial fill of the entry	SL quantity = filled quantity, raised as more fills arrive; the unfilled rest is cancelled after the timeout.
Order rejected (price band, margin, freeze qty)	Shown on the ticket with FYERS' message; no retry loop; trade -> CANCELLED or ALERT.
Two sells could fill	Impossible by design: only one exit order exists per trade.
You also trade manually in the FYERS app	The engine matches positions by symbol; a position it did not open is shown but never touched.
Duplicate clicks / double Arm	Every order carries a unique tag; the same tag is never sent twice.
Max loss per day reached	All Arm buttons disabled for the day; open trades keep their stops.
Kill switch	One button: cancel all working entries, exit all trades with limit orders.
15:10	Everything is closed with limit orders (your rule).
Expiry day	Warning on the ticket (fast premium decay, wide spreads); optional block after a time you set.

Situation	What the system does
Real orders	Only when the Real switch is ON and today is armed (python arm_real_orders.py) and your static IP is whitelisted at FYERS (SEBI rule). Otherwise everything runs as paper with the same screens.

9. Code structure (branch feature/chart-risk-engine)

New files next to what already exists; the BBP_NM strategy and the backtest are not changed.

```
kunal-fyers-terminal/
terminal/risk/sizing.py dual-constraint quantity (plain / strict), charges
terminal/risk/levels.py box -> validated plan (tick rounding, R:R, checks)
terminal/risk/trade.py trade state machine + saving / restoring
terminal/risk/orders.py place / modify / cancel via FYERS, one-exit-order OCO
terminal/risk/stream.py FYERS data + order websockets (ticks, fills)
terminal/risk/guard.py daily loss, max trades, kill switch, 15:10, watchdog
terminal/risk/server.py local link chart <-> engine (only on this PC)
static/risk_box.js the Long/Short box drawing, handles, ticket
app.py new view 'Manual trade (R:R box)'
tests/test_risk_*.py every rule above as a test, fake FYERS
Reused: fyers.py (login, quotes, contract list), live_trading/ (limit orders, arm switch,
journal), chart, settings, backtest.
```

10. Build plan

Phase	What you get	How it is checked
1. Sizing + ticket	Draw a box on the option chart, see lots / risk / reward live	tests with your examples (SENSEX 20: 10 lots plain, 8 strict)
2. Paper execution	Arm, entry, SL, target, drag-modify, exit - all simulated on live prices	a full paper day; journal shows every would-be order
3. Real orders, 1 lot	Same screens, real FYERS orders, 1 lot only	you watch the FYERS orderbook change as you drag
4. Full size + index chart	Normal sizing, boxes on the index chart, disaster stop	1 week live with small risk

11. Your day with it

- Folder C:\Users\USER\Desktop\Kunal\kunal-fyers-terminal: double-click start_terminal.bat, log in.
- Open 'Manual trade (R:R box)'. Set (once): capital allocation %, max risk per trade, max loss per day, buffer, strict sizing.
- Pick the index and the option (or click a strike in the option chain): its chart opens.
- Click 'Long box', click your entry, stop and target on the chart. Read the ticket.
- Press Arm. Watch: entry working -> in trade (green) -> target / stop. Drag the stop to trail.
- Real money only after: Settings 'Real orders' ON + python arm_real_orders.py (today) + static IP.

12. Questions for you (answers decide the details)

- Q1 Is drawing the box on the terminal chart (not FYERS Web) OK for you?
- Q2 Phase 1 on the OPTION chart only - OK? Later: which strike rule for boxes on the index chart (ATM / ATM+1 / premium range / per lot amount)?
- Q3 Sizing: Strict (worst fill + charges, recommended) or Plain (exactly your formula)?
- Q4 Entry: only when the price reaches the box entry (pending), or also 'enter now at CMP + 2'?
- Q5 Default: stop at the exchange + target watched by the engine (safer) - or target at the exchange + stop watched?
- Q6 Dragging the stop further away so that risk goes above Max risk: refuse the drag, or reduce quantity automatically?
- Q7 Several trades at the same time? Max how many, and max loss per day in rupees?
- Q8 Partial exits (e.g. half at 1:1, rest trailed) - wanted now or later?
- Q9 Should the existing BBP_NM auto-strategy and the manual box trades share one capital pool?
- Q10 Is your static IP already whitelisted at FYERS for API orders?

13. Your answers (30 Oct 2026) and what was built

Question	Your answer	Built as
Q1 terminal chart	OK if there is no lag (milliseconds fine, not seconds)	FYERS live websocket -> desk -> chart over a local link: about 1 ms on this PC (measured in the tests); an order change = one FYERS call (about 100-300 ms, FYERS side)
Q2 option chart first	agreed	boxes on the option chart; index-chart boxes later
Q3 sizing	your formula by default + a switch	PLAIN default (10 lots in the SENSEX example); 'Strict sizing' switch (8 lots)
Kill switch	on every page; broker kill switch; no new orders for 12 h	red KILL SWITCH on every page: cancel all + limit close-out + 12 h lock. FYERS has no API for its own kill switch, so the screen tells you to press 'Take a Break' in FYERS
Q4 entry	only when the price reaches the box	ARMED until reached, then BUY LIMIT price + 2
Q5 exits	the safer one	SL-L at FYERS, target watched, same order modified
Q6 risky drag	do not reduce the quantity	the drag is refused and the line jumps back
Q7 trades / loss	no cap; max loss box	no trade cap; 'Max loss per day' box (empty = none)
Q8 partial exits	later	not built
Q9 capital pool	same pool later, separate now	desk has its own capital for now
Q10 static IP	to be confirmed	real orders stay OFF until you confirm; then 1 lot first

Notes: this is a tool that executes YOUR decisions; it is not advice and cannot remove market risk (gaps, fast moves, liquidity). SEBI's 2026 rules for retail API trading (static IP, order rate limits, algo tagging) apply - FYERS' current requirements must be checked before going live with real orders.